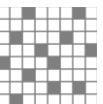


V4390

Controller Board

Document Revision: 1.2

Technical Reference

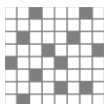


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Revision history

Revision	Description	Date
1.0	Initial Release	2016-06-15
1.1	Add DLP6500	2017-11-30
1.2	Review section DMD Assembly .	2020-09-11

Introduction

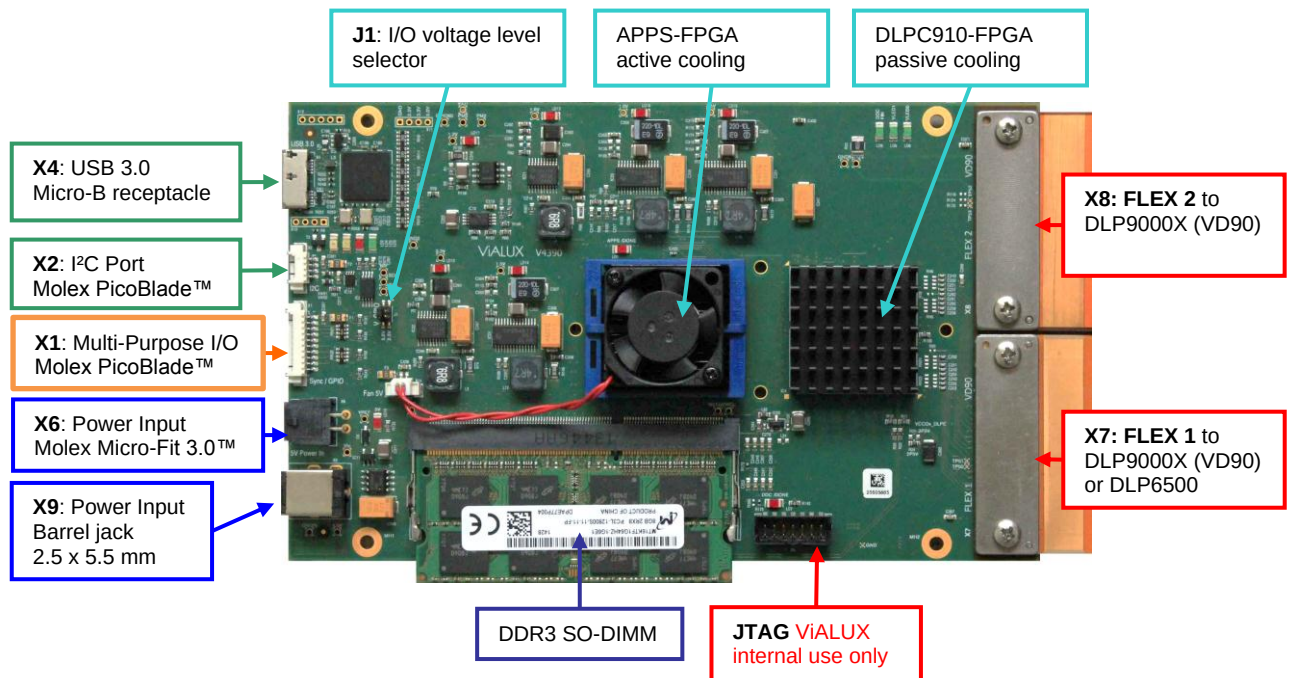
This document describes how to connect and operate the ViALUX V4390 DLP® Controller Board of SuperSpeed V-Modules V-9001 and V-6501.

The DLP® V4390 Controller Board provides control logic and power supply for driving DLP9000X DMDs (WQXGA 2560x1600) and DLP6500 DMDs (1920x1080) using Texas Instruments DLPC910 chipset.

Key features of the V4390 Controller Board are

- USB 3.0 SuperSpeed PC interface
- 8GB DDR3 SDRAM Memory on board

Connectors



SO-DIMM socket

A standard DDR3 SO-DIMM (small outline dual-inline memory module, 204 pins) is used for image storage. Please use the module specified and supplied by ViALUX only. Always switch off the device before removal or insertion of the memory module.

Power connectors

A 4-pin Molex DC power jack or, alternatively, a Barrel Receptacle is provided on the V4390 to connect to a power supply unit. Please do **not** use both power connectors simultaneously.

X6: Molex Micro-Fit 3.0™



This is a Molex Micro-Fit 3.0™ dual row header, Molex part number 43045-0412.

It mates with receptacle 43025-04xx.

Pin Number	Signal Name	Power Consumption
3 – 4 (upper pins)	5 V	6 A
1 – 2 (lower pins)	GND	

X9: DC Power Barrel Jack



Center pin diameter 2.5 mm, outside diameter 5.5 mm, Insertion Depth 9 mm

Pin	Signal Name	Power Consumption
Inside	5 V	6 A
Outside	GND	

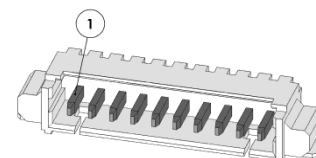
Use the barrel connector alternatively only if X6 is not in use. For production environment we recommend usage of X6 due to its interlock mechanism.

X4: USB 3.0 Micro-B connector



This is the connector for USB connection to the PC. Use the supplied USB cable to connect the board.

X1: Multi-Purpose I/O connector



The connector is a Molex PicoBlade™ header, part number 53261-1071. It mates with Molex part number 51021-1000. Use crimp contacts 50058-8000 or 50079-8000.

For your convenience the package includes a set of prepared cables and a connector housing (see picture left).

This connector is not hot-plug capable. Do not attach or detach the plug while the device is powered in order to avoid damage. Always apply a common GND when connecting to other devices.

The table below gives default directions and usage of pins after ALP-4 API initialization.

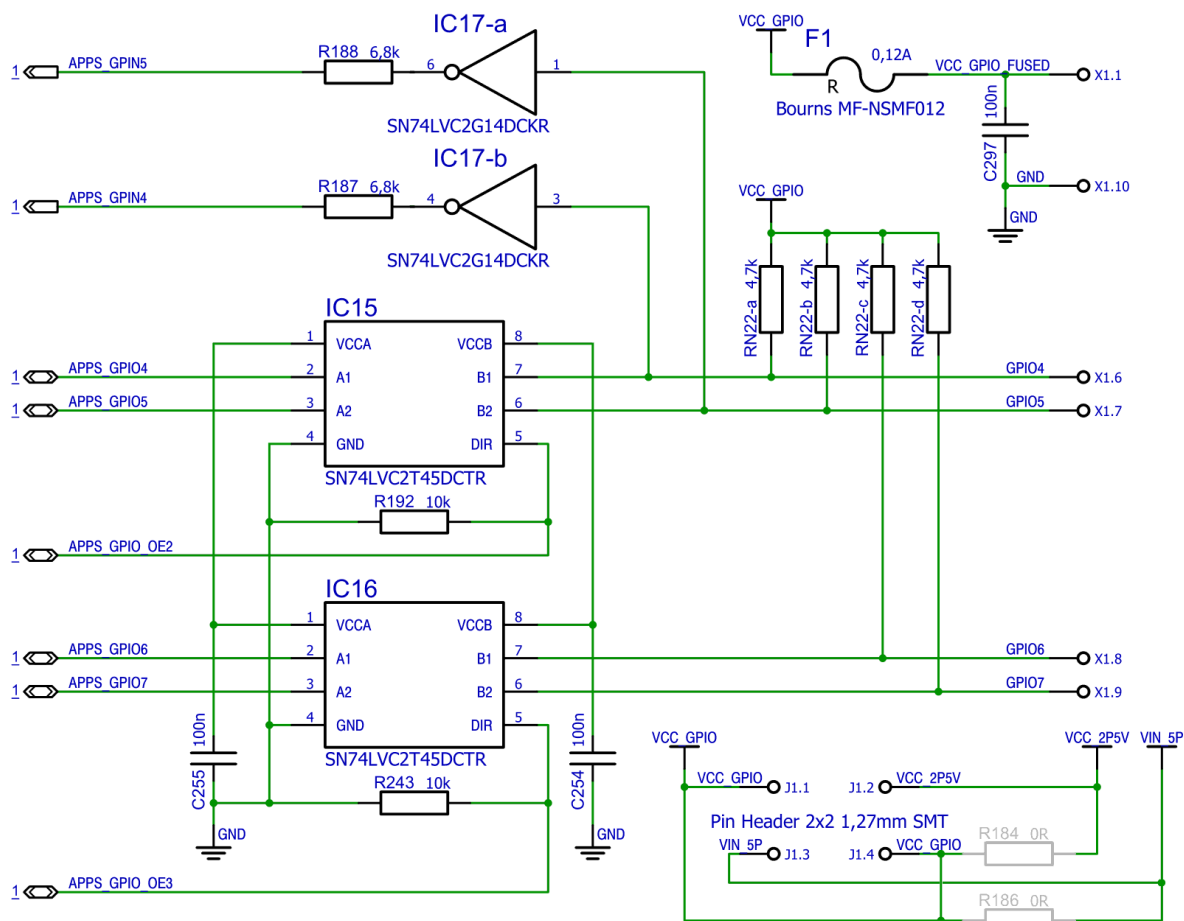
The ALP-4 API can change signal direction for whole groups only (A – D). For details and other usage please refer ALP-4 API description.

On power-up, all signal pins (2 to 9) are configured as inputs.

Pin	Signal Name	Direction	Group	Purpose
1	VCC_GPIO	IN/OUT		I/O Voltage (see configuration of jumper J1)
2	SYNCH_OUT1_GATE	OUT	A	Dynamic Gate Synch Output 1, refer to the ALP-4 API description, ALP_DEV_DYN_SYNCH_OUT1_GATE
3	SYNCH_OUT2_GATE	OUT	A	ALP_DEV_DYN_SYNCH_OUT2_GATE
4	SYNCH_OUT3_GATE	OUT	B	ALP_DEV_DYN_SYNCH_OUT3_GATE
5		OUT	B	Reserved for future use
6		IN	C	Reserved for future use
7	TRIGGER_IN	IN	C	Trigger Input
8	SYNCH_OUT	OUT	D	Frame Synchronization Output
9	PWM	OUT	D	ALP_PWM_LEVEL: output a pulse-width modulated value
10	GND	–		Power ground

Weak pull-up resistors (4k7 to VCC_GPIO) on all signals provide a defined level for each pin when floating (input). In order to satisfy operating conditions, do not add external pull-down resistors. For details refer to the data sheet of SN74LVC2T45 (pins 2 to 9; Texas Instrument, ti.com) and the schematic below.

Additional Schmitt Triggers SN74LVC2G14 at pins 6 and 7 improve robustness when configured as input.



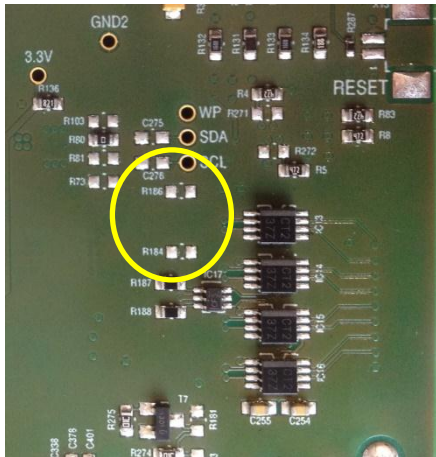
Jumper J1: I/O voltage selection

Jumper J1 has 1.27 mm pitch and is used to select either 2.5 V or 5.0 V voltage supply for the Multi-Purpose I/O connector.

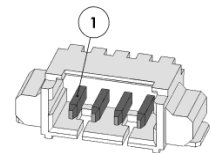
The attached hardware can supply the voltage level, when the jumper is left open. VCC_GPIO is connected to pin 1 of X1 (Multi-Purpose I/O) via a resettable fuse.

Position	0R (0603)	I/O voltage	VCC_GPIO source
open	-	2.5 V to 5.0 V	external (X1.1)
1 – 2	R184	2.5 V	Internal
3 – 4	R186	5.0 V	Internal

For use in high vibrational environments it is recommended to substitute the jumper by a single 0R resistor (size: 0603). Positions R184 and R186 are located on bottom side below J1:



Always use one jumper only or alternatively one single resistor in order to avoid board damage by a short circuit between both VCC sources.



X2: I2C connector

This connector is a Molex PicoBlade™ header, part number 53261-0471. It mates with Molex part number 51021-0400, use crimp contacts 50058-8000 or 50079-8000.

This connector is not hot-plug capable. Do not attach or detach a plug while the device is powered in order to avoid damage. Always apply a common GND when connecting devices.

Pin	Signal Name	Direction	Purpose
1	GND		Power ground
2	VCC_3P3V	OUT	Power (3.3V, max. 120mA, fused)
3	SDA	IN/OUT	Serial Data
4	SCL	OUT	Serial Clock

LED description

Table 1 lists the on board LEDs with its functional description.

Table 1 – Description of LEDs

LED	Color	Status	Description
LD4	Yellow	Steady	USB connected
		Blinks	USB active transferring control/data
LD2	Blue	Steady	USB link is in HighSpeed/SuperSpeed mode
LD5	Red	Steady	USB link is in SuperSpeed mode: LD4 and LD2 are also on; USB suspended: LD2 and LD4 off
LD3	Green	Steady	APPS FINISHED: FPGA initialization completed
LD1	Red	Steady	APPS /DONE: FPGA not configured
LD6	Green	Steady	DDC FIN: FPGA initialization completed
LD7	Red	Steady	DDC /DONE: FPGA not configured
LD8	Green	Blinks	VLED0: System “heartbeat” signal (1 Hz, 50% duty cycle)
LD9	Green	Steady	VLED1: denotes initialization complete
LD10	Red	Steady	Power Input 5V/6A (X6, X9) connected and working
LD11	Red	Steady	On-board 1.2 V power regulator output
LD12	Red	Steady	On-board 3.3 V power regulator output
LD13	Red	Steady	On-board 1.8 V power regulator output
LD14	Red	Steady	On-board 1.5 V power regulator output
LD15	Red	Steady	On-board 2.5 V power regulator output
LD16	Red	Steady	On-board 1.0 V power regulator output

Protection features

The V4390 board provides protection logic control for the DMD as well as for input voltage and polarity.

Power float

In the event of power failure or if the 5 V power supply is shut down, the DMD will be issued a FLOAT command that places all mirrors in a non-deflected position. The threshold is at 4.5 V. The ALP-4 API implements ALP_DEV_DMD_MODE for checking this state.

Avoid long power supply cables in order to prevent transient voltage drop below this threshold.

Operating conditions

Ambient conditions		
Storage temperature	-40 ... 80 °C	
Relative humidity	0 ... 95 % (non-condensing)	
DMD operating conditions	see DLP9000 data sheet (ti.com) and DLP6500 (ti.com)	
Power supply		
Voltage	Minimum 4.8 V, Typical 5 V, Absolute Maximum 6 V	
Current	Maximum 6 A	
Maximum temperature reading		
The V4390 contains three temperature measurement points that can be read out by the ALP API. One point is located near the junction of each FPGA (APPS-FPGA, DDC-FPGA), and one is on board, below the SO-DIMM. The device is designed to operate at the following temperatures. Note that exceeding this range could result in unreliable operation or device damage.		
Parameter	Value	ALP API InquireType
Board operating temperature:	10 ... 70 °C	ALP_PCB_TEMPERATURE
DDC-FPGA	≤ 80 °C	ALP_DDC_FPGA_TEMPERATURE
APPS-FPGA	≤ 80 °C	ALP_APPS_FPGA_TEMPERATURE
DMD temperature	Application specific	No on-board sensor available

Cooling

On the V4390 controller board application FPGA requires active cooling while for DLPC910 FPGA cooling a passive solution fits common requirements. As an option, V4390 controller board can be delivered without cooling if you intend to apply your own cooling solutions.

DMD Assembly

The DMD is mounted by pressing it into a contact-spring array, so it can be exchanged without soldering. The mechanical force (screw torque) shall be well controlled as given in the table.

Three A-Type DMDs in two sizes of housing are supported by the V4390 controller board. The table below refers to the 4x M3 screws in the original mounting hardware of the corresponding V-Modules.

V-Module Type	DMD Extension Board	Torque Specification for V-Module Mounting Set
V-9001	VD90	0.45 ± 0.11 Nm (4.0 ± 1.0 IN·LB)
V-9001 UV		
V-6501	Tbd	0.34 ± 0.11 Nm (3.0 ± 1.0 IN·LB)

Warning: Custom DMD mounting design requires careful consideration of the mechanical forces applied to the DMD. The DMD data sheets specify respective limits in section “Systems Mounting Interface Loads”.

Please see also <https://www.ti.com/product/DLP6500FLQ#design-development##design-tools-simulation> and <https://www.ti.com/product/DLP9000X#design-development##design-tools-simulation> for understanding the mounting concepts.

Flex Connections

The DMD Extension board VD90 is intended to connect V4390 controller board using two flex cable assemblies (FLEX1+FLEX2). For DLP6500 only one flex cable is required (FLEX1). Two lengths are available:

Part	ViALUX Order Number
Flex assembly with bolster plates, length 5.2”, 150 pos.	K-0104
Flex assembly with bolster plates, length 12.2”, 150 pos.	K-0105

Mounting Instructions

Observe ESD requirements strictly!

The V4390 Board is designed to work in any mounting position. Good air circulation is recommended.

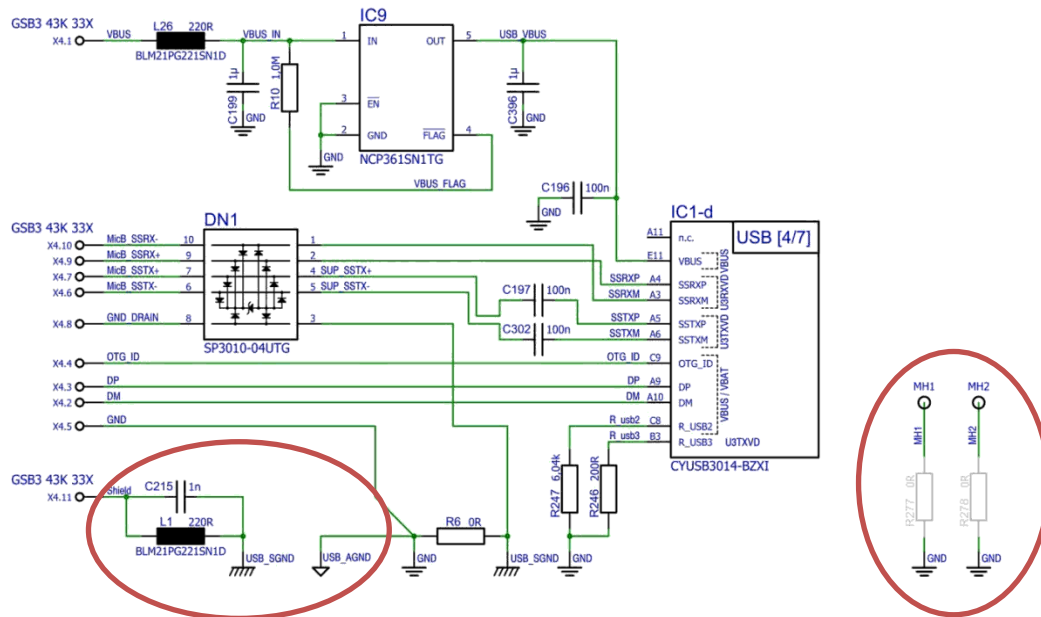
Never connect DLP650LNIR, DLP7000, DLP9500, or DLP9600 DMDs to a V4390 controller board!

Ground and Shield Concept

All mounting holes of the V4390 Main Board are potential free and therefore not grounded or shielded. GND is connected to the power supply input (X6: Pin 1-2, X9: outside).

USB input (X4) is handled as follows. GND is connected to the USB-Ground directly, where shielding has a weak AC-connection and a short DC-connection using C215 and L1 (see figure below).

Usually V4390 Main Board is driven by a single 5V/6A power supply with GND connection to the safety contact of the 100V...240V AC wall sockets.



Dimension Drawing

See also [V4390 Dimensions.pdf](#)